Research Notes on Gaussian process

Gaussian process

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

>> Section 1

and maximizing this marginal likelihood towards θ provides the complete specification of the Gaussian process f. One can briefly note at this point that the first term corresponds to a penalty term for a model's failure to fit observed values and the second term to a penalty term that increases proportionally to a model's complexity. Having specified θ, making predictions about unobserved values ■ f (x ∗) {\displaystyle f(x^{*})} ■ at coordinates x* is then only a matter of drawing samples from the predictive distribution p (y ∗ ■ x ∗ , f (x) , x) = N (y ∗ ■ A , B) {\displaystyle p(y^{*}\mid x^{*},f(x),x)=N(y^{*}\mid A,B)} where the posterior mean estimate A is defined as

>> Section 2

Definition A time continuous stochastic process { X t ; t ∈ T } {\displaystyle \left\{X_{t};t\in T\right\}} is Gaussian if and only if for every finite set of indices t 1 , … , t k {\displaystyle t_{1},\ldots ,t_{k}} in the index set T {\displaystyle T}

>> Section 3

Literature The Gaussian Processes Web Site, including the text of Rasmussen and Williams' Gaussian Processes for Machine Learning Ebden, Mark (2015). "Gaussian Processes: A Quick Introduction". arXiv:1505.02965 [math.ST]. A Review of Gaussian Random Fields and Correlation Functions Efficient Reinforcement Learning using Gaussian Processes

>> Section 4

Physical applications Gaussian processes have found increasing applications in many domains of the natural sciences due to their statistical modelling properties. Molecular property prediction has employed these process models in small molecular datasets due to their inference capabilities and computational costs. They are also being increasingly used as surrogate models for force field optimization.

>> Section 5

lim n → ∞ tr ■ [R n R n − 1] = lim n → ∞ tr ■ [I] = lim n → ∞ n = ∞ . {\displaystyle \lim _{n\to \infty }\operatorname {tr} [R_{n}R_{n}^{-1}]=\lim _{n\to \infty }\operatorname {tr} [I]=\lim _{n\to \infty }n=\infty .}

>> Section 6

In practical applications, Gaussian process models are often evaluated on a grid leading to multivariate normal distributions. Using these models for prediction or parameter estimation using maximum likelihood requires evaluating a multivariate Gaussian density, which involves calculating the determinant and the inverse of the covariance matrix. Both of these operations have cubic computational complexity which means that even for grids of modest sizes, both operations can have a prohibitive computational cost. This drawback led to the development of multiple approximation methods.

>> Section 7

or E [e i s (X t − μ)] = e − s σ s / 2 {\displaystyle {\mathbb {E} }\left[{\mathrm {e} }^{i\,\mathbf {s} \,(\mathbf {X} _{t}-\mathbf {\mu })}\right]={\mathrm {e} }^{-\mathbf {s} \,\sigma \,\mathbf {s} /2}} . The numbers σ ■ j {\displaystyle \sigma _{\ell j}} and μ ■ {\displaystyle \mu _{\ell }} can be shown to be the covariances and means of the variables in the process.

>> Section 8

E ∑ n c n (| ξ n | + | η n |) = ∑ n c n E [| ξ n | + | η n |] = const ⋅ ∑ n c n < ∞ , {\textstyle {\mathbb {E} }\sum _{n}c_{n}(|\xi _{n}|+|\eta _{n}|)=\sum _{n}c_{n}{\mathbb {E} }[|\xi _{n}|+|\eta _{n}|]={\text{const}}\cdot \sum _{n}c_{n}<\infty ,}

>> Section 9

does not follow from continuity of σ {\displaystyle \sigma } and the evident relations σ (h) ≥ 0 {\displaystyle \sigma (h)\geq 0} (for all h {\displaystyle h}) and σ (0) = 0. {\displaystyle \sigma (0)=0.}

>> Section 10

A Gaussian process can be used as a prior probability distribution over functions in Bayesian inference. Given any set of N points in the desired domain of the functions, take a multivariate Gaussian whose covariance matrix parameter is the Gram matrix of those N points with some desired kernel, and sample from that Gaussian. For solution of the multi-output prediction problem, Gaussian process regression for vector-valued function was developed. In this method, a 'big' covariance is constructed, which describes the correlations between all the input and output variables taken in N points in the desired domain. This approach was elaborated in detail for the matrix-valued Gaussian processes and generalised to processes with 'heavier tails' like Student-t processes. Inference of continuous values with a Gaussian process prior is known as Gaussian process regression, or kriging; extending Gaussian process regression to multiple target variables is known as cokriging. Gaussian processes are thus useful as a powerful non-linear multivariate interpolation tool. Kriging is also used to extend Gaussian process in the case of mixed integer inputs. Gaussian processes are also commonly used to tackle numerical analysis problems such as numerical integration, solving differential equations, or optimisation in the field of probabilistic numerics. Gaussian processes can also be used in the context of mixture of experts models, for example. The underlying rationale of such a learning framework consists in the assumption that a given mapping cannot be well captured by a single Gaussian process model. Instead, the observation space is divided into subsets, each of which is characterized by a different mapping function; each of these is learned via a different Gaussian process component in the postulated mixture.

>> Section 11

In probability theory and statistics, a Gaussian process is a stochastic process (a collection of random variables indexed by time or space), such that every finite collection of those random variables has a multivariate normal distribution. The distribution of a Gaussian process is the joint distribution of all those (infinitely many) random variables, and as such, it is a distribution over functions with a continuous domain, e.g. time or space. The concept of Gaussian processes is named after Carl Friedrich Gauss because it is based on the notion of the Gaussian distribution (normal distribution). Gaussian processes can be seen as an infinite-dimensional generalization of multivariate normal distributions. Gaussian processes are useful in statistical modelling, benefiting from properties inherited from the normal distribution. For example, if a random process is modelled as a Gaussian process, the distributions of various derived quantities can be obtained explicitly. Such quantities include the average value of the process over a range of times and the error in estimating the average using sample values at a small set of times. While exact models often scale poorly as the amount of data increases, multiple approximation methods have been developed which often retain good accuracy while drastically reducing computation time.